

Lexical Analyzer in NLP

What is a Lexical Analyzer?

- Converts sentences into tokens
- Identifies words, numbers, and symbols
- Supports language-independent preprocessing
- Forms the foundation for syntax analysis
- Works with text normalization
- Example: 'Hello World' → Hello, World

Key Functions

- Tokenization of text
- Removal of stop words
- Case normalization (lowercasing)
- Handling punctuation and symbols
- Basic error detection in text
- Creates structured token output

Working Process

- Receive raw text as input
- Split text into individual tokens
- Normalize words and characters
- Filter unwanted tokens
- Generate clean token sequence
- Pass output to next NLP stage
- Example: 'Cats are running.' → cats, running

Applications & Benefits

- Used in chatbots and virtual assistants
- Supports sentiment analysis
- Improves information retrieval
- Essential for machine translation
- Enhances text classification accuracy
- Reduces preprocessing time

Conclusion

- Lexical analysis is the first step in NLP text processing.
- It converts raw text into meaningful tokens.
- Proper tokenization improves NLP model performance.
- It enables accurate syntax and semantic analysis.
- Widely used in chatbots, search engines, and text analytics.
- A strong lexical analyzer builds the foundation for efficient NLP systems.

THANK YOU